

Monitoring Network Usage SigCap: Workflow, and Needs

How is Data Collected: SigCap, an Android App, Dr. Muhmmad walks around with a phone to collect data, and analyze it (not realtime).

Original Proposal: Automate the data process so there are no “by hand observations”

After Discussion: There is currently no need for any more automation, Dr. Ghosh is happy with the current flow, and data processing and uploads are already automatic.

Sigcap Info Vs Original Proposal:

Beacon: Data every 100ms

Beacon: Max Channel > 80MHz

RH Wifi: Also can check for a preamble

Wireshark: uses Wifi & “sniffs” it (Have a demo with a raspberry pi chip @ 80Mhz)

Sigcap can see how many access points per frequency

Channel Utilization:

Stadium Dense Dataset: 900 - 1400 access points: big thing is analyzing them

SigCap does not get any raw RF data no, IQ, instead the on phone chip / API handles it all and they recipe the wifi info (ie signal strength)

Ideas From Dr.Ghosh:

We could try downloading SigCap for us, and taking it to a football game (Possible up to 320MHz).

Dr. Ghosh wants to sense the wider channel, the full 320 MHz, currently SigCap is capped at 80MHz.

If we could decode a preamble, for Real Time CSI, or see how long it takes to wait and listen to a signal in real time.

We want Channel State info detecting it from wifi ex can be on an SDR or using legacy preambles.

Real time interference detection would be nice, but this would be extremely hard.

*Advised not to rush and go straight into data collection / IQ sampling of RH > Grab IQ sample in air, store, use in python > then do the signal processing

Suggested Topics From Dr.Ghosh:

Technical: How wifi waves don't bump into each other in air

New Proposal: Use RH for two main goals BLE, and Wifi detection.

Bluetooth: Currently they can only detect the # of bluetooth devices (thats it!). Can we get more information about BLE (example what frequency each device is on) with the goal of understanding Channel Occupancy when it comes to BLE VS Wifi and WIFI is being impaired by BLE (Think CSI Utilization).

WiFi: Detects more Channel Stage Info from Wifi. Right now SigCap sees the Channel Utilization %,

Ultimate Goal: Gaining a better understanding of WiFi and Bluetooth to manage it better.

Emails: S: Chris

Just throwing this thought your way. There's a method of signal sampling called "Equivalent Time Sampling" or ETS. It's a sub-Nyquist sampling strategy based on taking advantage of signals that are repetitive over time. (Such as WiFi's STF repeating a few times) Makes me wonder if we should and can make a WiFi signal detector dirt cheap ADC.

I'm not saying we have to pursue it immediately, but figured it would give you [and Timi and Yankai] something to think about and chew on. I'm attaching a Tektronix link that has a decent little graphic describing it.

(<https://www.tek.com/en/documents/application-note/real-time-versus-equivalent-time-sampling>)